

SMART CONTRACT SECURITY AUDIT - TOKENORACLEDEX

One-Page Visual Audit Report

Purpose: Toy DEX that buys/sells internal balances using a mutable price variable.

Risk	Severity	SWC	OWASP
setPrice() is unrestricted; anyone can manipulate the oracle price	Critical	SWC-105	SC03 Oracle Manipulation
sell() makes external call before balance reduction	High	SWC-107	SC08 Reentrancy
price can be zero, causing sell() division-by-zero DoS	Medium	N/A	SC05 Input Validation
Internal balances are not backed by robust token or liquidity accounting	High	SWC-132	SC02 Financial Risks

Root Cause: The DEX trusts a public mutable price and performs ETH transfers before finalizing internal accounting.

Business Logic Findings: Attackers can set favorable prices, mint cheap internal balance, and attempt to drain ETH liquidity. Zero price can break redemptions.

BENCHMARK METRICS

Time_seconds: 75 | Confidence: 0.97 | Total_findings: 4
Critical: 1 | High: 2 | Medium: 1 | Low: 0
Precision_estimate: 0.95 | Coverage_estimate: 1.00 | Risk_score: 95/100

GOLD STANDARD EVALUATION

TP: N/A | FP: N/A | FN: N/A | Precision: N/A | Recall: N/A | F1: N/A